Master Summary Sheet

Cohomology on $\mathbb{R}^2 \setminus \{0\}$, S^2, T^2, and Gauss–Bonnet

Main principle.

Cohomology measures when a closed form is not exact.

Equivalently:

local primitives may exist, but a global primitive may fail to exist.

1. de Rham cohomology

$$H^k_{\mathrm{dR}}(M) = \frac{\{\text{closed } k\text{-forms}\}}{\{\text{exact } k\text{-forms}\}}.$$

- ω is **closed** if $d\omega = 0$.
- ω is **exact** if $\omega = d\eta$.
- Exact $\Rightarrow$ closed, since $d^2 = 0$.
- To prove ω is *not* exact, find a closed cycle C with

$$\int_C \omega \neq 0.$$

2. Model example: winding form on $\mathbb{R}^2 \setminus \{0\}$

$$\eta = \frac{-y\,dx + x\,dy}{x^2 + y^2}.$$

In polar coordinates,

$$x = r\cos\theta, \qquad y = r\sin\theta,$$

one gets

$$\eta = d\theta.$$

So:

$$d\eta = 0.$$

But around the unit circle,

$$\int_{|z|=1} \eta = \int_0^{2\pi} d\theta = 2\pi \neq 0.$$

Therefore:

$$[\eta] \neq 0 \in H^1_{\mathrm{dR}}(\mathbb{R}^2 \setminus \{0\}).$$

Meaning. The angle θ exists locally but not as a globally single-valued function.

3. Sphere S^2

Using spherical coordinates,

$$X(\theta, \phi) = (\sin\phi\cos\theta, \sin\phi\sin\theta, \cos\phi),$$

the metric is

$$ds^2 = d\phi^2 + \sin^2\phi\, d\theta^2,$$

so the area form is

$$dA = \sin\phi\, d\theta \wedge d\phi.$$

Since S^2 is 2-dimensional,

$$d(dA) = 0,$$

so dA is closed.

Also,

$$\int_{S^2} dA = \int_0^{2\pi}\int_0^{\pi} \sin\phi\, d\phi\, d\theta = 4\pi.$$

Hence dA is not exact. Therefore

$$[dA] \neq 0 \in H^2_{\mathrm{dR}}(S^2).$$

So:

$$H^0_{\mathrm{dR}}(S^2) \cong \mathbb{R}, \qquad H^1_{\mathrm{dR}}(S^2) = 0, \qquad H^2_{\mathrm{dR}}(S^2) \cong \mathbb{R}.$$

Local primitives on S^2

On

$$U_N = S^2 \setminus \{N\}, \qquad U_S = S^2 \setminus \{S\},$$

define

$$\alpha_N = (1 - \cos\phi)\, d\theta, \qquad \alpha_S = -(1 + \cos\phi)\, d\theta.$$

Then locally,

$$d\alpha_N = dA, \qquad d\alpha_S = dA$$

up to orientation convention, and on the overlap,

$$\alpha_N - \alpha_S = 2\, d\theta.$$

Meaning. The area form is locally exact, but the local primitives differ by winding on the overlap.

4. Torus $T^2 = \mathbb{R}^2/\mathbb{Z}^2$

The forms dx, dy descend to global 1-forms on T^2, with

$$d(dx) = 0, \qquad d(dy) = 0.$$

Let

$$\gamma_x(t) = (t, 0) \mod \mathbb{Z}^2, \qquad \gamma_y(t) = (0, t) \mod \mathbb{Z}^2, \qquad 0 \leq t \leq 1.$$

Then

$$\int_{\gamma_x} dx = 1, \qquad \int_{\gamma_y} dy = 1,$$

so neither dx nor dy is exact.

Thus

$$[dx], [dy] \neq 0 \in H^1_{\mathrm{dR}}(T^2).$$

Now

$$dx \wedge dy$$

is closed, and

$$\int_{T^2} dx \wedge dy = 1,$$

so it is not exact. Hence

$$[dx \wedge dy] \neq 0 \in H^2_{\mathrm{dR}}(T^2).$$

Therefore

$$H^0_{\mathrm{dR}}(T^2) \cong \mathbb{R}, \qquad H^1_{\mathrm{dR}}(T^2) \cong \mathbb{R}^2, \qquad H^2_{\mathrm{dR}}(T^2) \cong \mathbb{R}.$$

The ring structure is

$$[dx] \wedge [dy] = [dx \wedge dy], \qquad [dy] \wedge [dx] = -[dx \wedge dy].$$

Meaning. On the torus, the two basic winding directions are $[dx]$ and $[dy]$, and their wedge gives the top class.

5. Curvature and Gauss–Bonnet

For an oriented Riemannian surface M, with Gaussian curvature K and area form dA, the curvature 2-form is

$$K\, dA.$$

Since M is 2-dimensional,

$$d(K\, dA) = 0,$$

so $K\, dA$ is closed.

Locally, if ω_{12} is the connection 1-form for a positively oriented orthonormal frame, then

$$d\omega_{12} = K\, dA.$$

So curvature is *locally exact*.

Gauss–Bonnet says:

$$\boxed{\int_M K\, dA = 2\pi\chi(M).}$$

Equivalently,

$$\boxed{\frac{1}{2\pi}[K\, dA] = e(TM)}$$

in de Rham cohomology, where $e(TM)$ is the Euler class.

6. Check Gauss–Bonnet on S^2 and T^2

Sphere

For the unit sphere,

$$K \equiv 1.$$

So

$$K\, dA = dA, \qquad \int_{S^2} K\, dA = 4\pi.$$

Since

$$\chi(S^2) = 2,$$

we get

$$4\pi = 2\pi\chi(S^2).$$

Flat torus

For the flat torus,

$$K \equiv 0.$$

So

$$K\,dA = 0, \qquad \int_{T^2} K\,dA = 0.$$

Since

$$\chi(T^2) = 0,$$

we get

$$0 = 2\pi\chi(T^2).$$

Important distinction.

$$[dA] \neq 0 \text{ on } T^2, \qquad [K\,dA] = 0 \text{ for the flat torus.}$$

Gauss–Bonnet is about $K\,dA$, not about dA alone.

7. Sphere transition function and Euler class

In stereographic coordinates,

$$z = \frac{X + iY}{1 - Z}, \qquad w = \frac{X - iY}{1 + Z},$$

and on the overlap,

$$w = \frac{1}{z}.$$

So the tangent-bundle transition function is

$$\frac{dw}{dz} = -\frac{1}{z^2}.$$

Along the equator, $z = e^{i\theta}$, so

$$\frac{dw}{dz} = -e^{-2i\theta}.$$

This winds twice around S^1. Hence the tangent bundle of S^2 has degree 2, so

$$e(TS^2)[S^2] = 2.$$

This matches:

$$\chi(S^2) = 2, \qquad \frac{1}{2\pi}\int_{S^2} K\,dA = 2.$$

8. Final memory map

Space	Closed not exact form	Nonzero cohomology	Gauss–Bonnet
$\mathbb{R}^2 \setminus \{0\}$	$\eta = d\theta$	H^1	winding
S^2	dA	H^2	$\int K\,dA = 4\pi$
T^2	dx, dy, $dx \wedge dy$	H^1, H^2	$\int K\,dA = 0$ (flat)

Ultimate summary.

Winding form $\longrightarrow$ cohomology class $\longrightarrow$ overlap obstruction $\longrightarrow$ Euler class $\longrightarrow$ Gauss–Bonnet.